

Product Requirements Document (PRD)

Product Name

Enterprise BI, Analytics & Intelligence Platform

Version

v1.1 – Expanded Master PRD

Owner

Product, Data & Platform Engineering

Status

Proposed / Strategic Design (Detailed)

1. Product Vision & Strategy

1.1 Vision

The platform is designed to be a **full-scale Intelligence Operating System**, not merely a visualization tool. It enables organizations to systematically convert raw data into **trusted, governed, explainable intelligence** that supports monitoring, planning, forecasting, and decision-making across domains.

The platform explicitly supports the analytics maturity curve:

- * **Descriptive**: What happened?
- * **Diagnostic**: Why did it happen?
- * **Predictive**: What will happen?
- * **Prescriptive**: What should we do next?

1.2 Strategic Goals

- * Replace fragmented BI, reporting, and analytics workflows with a unified system
- * Enforce governance, metadata, and standards by default
- * Enable domain-specific intelligence (starting with healthcare, extensible to finance)
- * Support internal unlimited use and future external licensing

1.3 Design Principles

- * Metadata-first, schema-aware architecture
- * Manual, intentional creation of dashboards and reports
- * Explainability over black-box automation
- * Guardrails instead of unrestricted flexibility
- * Separation of data, insight, and decision layers

2. Scope Definition

2.1 In Scope

- * Data ingestion from structured and semi-structured sources
- * Metadata modeling and semantic abstraction
- * Dashboard and visualization creation
- * Report authoring and publishing
- * Predictive analytics and forecasting
- * Prescriptive recommendation generation
- * AI-assisted explanation and narrative generation
- * Multi-language support
- * Governance, audit, and compliance

2.2 Out of Scope (Phase 1)

- * Streaming / real-time analytics
- * User-written transformation scripts
- * Auto-publishing to public websites
- * Autonomous AI-driven decision execution

3. User Personas & Responsibilities

3.1 Personas

Data Engineer

- * Connects data sources
- * Ensures data quality and refresh
- * Manages schemas and pipelines

Analyst

- * Builds dashboards and reports
- * Designs indicators and visuals
- * Validates analytical outputs

Program / Business Manager

- * Consumes dashboards
- * Uses predictive and prescriptive outputs
- * Shares reports internally and externally

Executive / Leadership

- * Consumes high-level dashboards and compiled reports
- * Focuses on decisions, not data exploration

Superadmin

- * Manages templates, standards, and branding
- * Controls governance and permissions
- * Approves production deployments

4. Platform Architecture (Conceptual)

The platform is composed of modular but tightly integrated layers:

1. Data Integration Layer
2. Metadata & Semantic Layer
3. Analytics & Computation Layer
4. Visualization & Dashboard Layer
5. Reporting & Publishing Layer
6. Predictive & Prescriptive Intelligence Layer
7. LLM & Conversational Layer
8. Governance, Security & Deployment Layer

Each layer is independently extensible but governed centrally.

5. Data Integration Layer

5.1 Supported Sources

- * Relational databases (via JDBC / DBeaver-style connectors)
- * CSV and Excel uploads
- * APIs (REST)
- * FHIR servers (health domain)
- * Uploaded documents (PDFs, annual reports)

5.2 Ingestion Capabilities

- * Batch ingestion

- * Scheduled refresh
- * Manual uploads
- * Versioned datasets

5.3 Data Validation & Quality

- * Schema validation
- * Null and completeness checks
- * Basic anomaly detection
- * Mandatory metadata completeness checks

6. Metadata & Semantic Layer (Core Differentiator)

6.1 Purpose

To decouple raw data structures from analytical and visual usage, enabling reuse, consistency, and governance.

6.2 Metadata Types

- * Dataset metadata (source, refresh, ownership)
- * Indicator definitions (numerator, denominator, unit)
- * Dimensions (time, geography, demographics)
- * Domain ontologies (FHIR, ICD, SNOMED)
- * Governance tags (public, internal, restricted)

6.3 Semantic Models

- * Metrics defined once, reused everywhere
- * Version-controlled semantic definitions
- * Auditable change history

7. Dashboarding & Visualization Module

7.1 Dashboard Definition

A dashboard is a ****manually created analytical product**** with a defined purpose, owner, and audience.

Rules

- * Minimum pages: 1
- * Maximum pages: 9
- * Dashboard creation is never automatic

7.2 Page Architecture

Mandatory Page Types

- * Overview
- * Trends & Time
- * Geography (if spatial data exists)

Optional Page Types

- * Stratification
- * Program / Cascade
- * Predictive
- * Prescriptive
- * Data Quality

Each page answers exactly one primary question.

7.3 Charts & Layout Rules

- * Ideal charts per page: 4–6
- * Absolute maximum: 8
- * Grid-based (12-column) layout
- * Snap-to-grid resizing and repositioning
- * Responsive behavior across devices

7.4 Maps

- * Offline global base maps
- * Choropleth, heat, and point layers
- * Drill-down by geography

8. Reporting & Publishing Module

Reports are structured narrative documents derived from dashboards and datasets.

8.1 Report Capabilities

- * Rich text authoring
- * Embedded charts, tables, and maps
- * Inline citations and bibliographies
- * Multi-language variants
- * Branding and templates

8.2 Lifecycle

- * Draft → Published → Archived
- * Versioned with rollback

9. Predictive Analytics Engine

9.1 Purpose

To forecast future states and risks based on historical data.

9.2 Capabilities

- * Time-series forecasting
- * Trend and seasonality analysis

- * Scenario projections

9.3 Governance

- * Transparent assumptions
- * Confidence intervals
- * Clear labeling of forecasts vs observed data

10. Prescriptive Analytics Engine

10.1 Purpose

To recommend actions based on predictive insights and defined constraints.

10.2 Prescriptive Templates

Templates include:

- * Trigger conditions
- * Allowed interventions
- * Constraints (budget, capacity)
- * Expected outcomes

10.3 Outputs

- * Ranked recommendations

- * Rationale and assumptions
- * Scenario comparisons

11. LLM & Conversational Layer

11.1 Capabilities

- * Explain charts and dashboards
- * Summarize reports
- * Generate narrative drafts
- * Assist with translations

11.2 Restrictions

- * Cannot create or modify data
- * Cannot bypass governance
- * Cannot auto-publish outputs

12. Multi-Language & Accessibility

- * UI localization
- * Report language variants
- * Configurable fonts and sizes
- * Accessibility-friendly color schemes

13. Templates & Standardization

13.1 Template Types

- * Dashboard templates
- * Report templates
- * Prescriptive templates

13.2 Governance

- * Superadmin-only creation
- * Versioned templates
- * Mandatory templates for public outputs

14. Deployment & Environment Management

14.1 Environments

- * Development
- * UAT
- * Production

14.2 Promotion Rules

- * Manual promotion only
- * Approval workflow
- * Rollback supported

15. Security, Access & Permissions

- * Login-based authentication
- * Role-based access control
- * Dataset-, dashboard-, and report-level permissions

16. Audit, Compliance & Trust

- * Full activity logging
- * Data lineage tracking
- * Version history
- * Compliance-ready exports

17. Performance & Scalability

- * Page load targets < 3 seconds
- * Support for large datasets

- * Horizontal scalability

18. Productization & Licensing

- * Internal unlimited usage
- * Feature- and role-based licensing
- * Domain-specific add-ons

19. Success Metrics

- * User adoption
- * Time to insight
- * Reduction in manual reporting
- * User trust scores

20. Roadmap & Future Enhancements

- * Advanced ML models
- * Optimization engines
- * Real-time analytics
- * Public data portals

Final Product Statement

> This platform institutionalizes intelligence, ensuring that data-driven decisions are
repeatable, explainable, governed, and scalable across domains.